

| Question number | Answer                                                                                                                                                                                                                                                                                                           | Mark |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 6(a)(i)         | <p>The only correct answer is C</p> <p><i>A is incorrect because the base is not attached to the phosphate group</i><br/> <i>B is incorrect because the phosphate group is not attached to the base</i><br/> <i>D is incorrect because the phosphate group and base are not attached to adjacent carbons</i></p> | (1)  |

| Question number | Answer                                                                             | Additional guidance | Mark |
|-----------------|------------------------------------------------------------------------------------|---------------------|------|
| 6(a)(ii)        | <ul style="list-style-type: none"> <li>phosphodiester (bond / link) (1)</li> </ul> |                     | (1)  |

| Question number | Answer                                                                                                                                                                                                                                                                                                                                                                                                   | Additional guidance                                                                                                                                                                                                                                                                                                                                                    | Mark |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 6(b)(i)         | <p>An explanation that includes the following points:</p> <ul style="list-style-type: none"> <li>as temperature increases the {rate of enzyme reactions / length of lizards} increases (1)</li> <li>due to more {frequent collisions / energetic collisions} between enzymes and substrates (1)</li> <li>credit a named enzyme-controlled reaction that would contribute to faster growth (1)</li> </ul> | <p><b>ACCEPT</b> as rate of enzyme reactions increase the length of lizards increases<br/>increases enzyme activity</p> <p><b>ACCEPT</b> due to more enzyme substrate complexes forming<br/>due to increase in kinetic energy<br/>increase in {metabolic reactions / metabolism}</p> <p>e.g. DNA synthesis, mitosis, protein synthesis, respiration, cell division</p> | (3)  |

| Question number | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Additional guidance                                                                                                                                                                 | Mark |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| (b)(ii)         | <p>A description that includes two of the following points:</p> <ul style="list-style-type: none"> <li>• at 46 mm there is no change in the abundance of lizards (1)</li> <li>• as the length of the lizard increases above 46 mm there is decrease in the abundance of lizards (1)</li> <li>• lizards {between 44.5 and 46 mm / less than 46 mm} in length increased in abundance (1)</li> <li>• the length of the lizard affects its {chance of survival / abundance} (1)</li> </ul> | <p><b>ACCEPT</b> shorter lizards (are more likely to) have a greater chance of survival / converse<br/>lizards of 44.5 mm (are more likely to) have greatest chance of survival</p> | (2)  |

| Question number | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                | Additional guidance                                                 | Mark       |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|------------|
| 6(b)(iii)       | <p>An explanation that includes three of the following points:</p> <ul style="list-style-type: none"> <li>• because the cells in the longer lizards would have divided more (1)</li> <li>• and every time a cell divides DNA synthesis takes place (1)</li> <li>• {shorter telomeres / damaged chromosomes} cause cell death (1)</li> <li>• so the longer lizards are in lower abundance / lizards die before they can grow any longer (1)</li> </ul> | <p><b>ACCEPT</b> number of cell divisions increases with growth</p> | <p>(3)</p> |